



LLM-RecG: A Semantic Bias-Aware Framework for Zero-Shot Sequential Recommendation

Yunzhe Li

Siebel school of computing and data science
University of Illinois, Urbana-Champaign
CHAMPAIGN, IL, USA
yunzhel2@illinois.edu

Hari Sundaram

Siebel school of computing and data science
University of Illinois at Urbana-Champaign
CHAMPAIGN, IL, USA
hs1@illinois.edu

Junting Wang

Siebel school of computing and data science
University of Illinois at Urbana-Champaign
CHAMPAIGN, IL, USA
junting3@illinois.edu

Zhining Liu

Siebel school of computing and data science
University of Illinois at Urbana-Champaign
CHAMPAIGN, IL, USA
liu326@illinois.edu

Abstract

Zero-shot cross-domain sequential recommendation (ZCDSR) enables predictions in unseen domains without additional training or fine-tuning, addressing the limitations of traditional models in sparse data environments. Recent advancements in large language models (LLMs) have significantly enhanced ZCDSR by facilitating cross-domain knowledge transfer through rich, pretrained representations. Despite this progress, domain semantic bias—arising from differences in vocabulary and content focus between domains—remains a persistent challenge, leading to misaligned item embeddings and reduced generalization across domains.

To address this, we propose a novel semantic bias-aware framework that enhances LLM-based ZCDSR by improving cross-domain alignment at both the item and sequential levels. At the item level, we introduce a generalization loss that aligns the embeddings of items across domains (inter-domain compactness), while preserving the unique characteristics of each item within its own domain (intra-domain diversity). This ensures that item embeddings can be transferred effectively between domains without collapsing into overly generic or uniform representations. At the sequential level, we develop a method to transfer user behavioral patterns by clustering source domain user sequences and applying attention-based aggregation during target domain inference. We dynamically adapt user embeddings to unseen domains, enabling effective zero-shot recommendations without requiring target-domain interactions.

Extensive experiments across multiple datasets and domains demonstrate that our framework significantly enhances the performance of sequential recommendation models on the ZCDSR task. By addressing domain bias and improving the transfer of sequential patterns, our method offers a scalable and robust solution for better knowledge transfer, enabling improved zero-shot recommendations across domains.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Sequential Recommendation, Zero-Shot Transfer, Large Language Models, Pre-trained Model

ACM Reference Format:

Yunzhe Li, Junting Wang, Hari Sundaram, and Zhining Liu. 2025. LLM-RecG: A Semantic Bias-Aware Framework for Zero-Shot Sequential Recommendation. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25)*, September 22–26, 2025, Prague, Czech Republic. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3705328.3748077>

1 Introduction

Zero-shot cross-domain sequential recommendation (ZCDSR) extends beyond conventional recommendation tasks [17, 31] by addressing the critical challenge of making accurate predictions in unseen domains where no prior interaction data exists. This capability is indispensable in dynamic environments—such as new markets, product categories, or emerging user segments—where collecting domain-specific interaction data is often impractical, and it is particularly relevant in modern applications like e-commerce, streaming platforms, and online education, which frequently encounter rapid changes and the introduction of new domains. Traditional models, which rely heavily on domain-specific training data, often struggle to adapt in such scenarios, resulting in poor recommendation quality and a diminished user experience. ZCDSR plays a crucial role in overcoming this limitation by enabling recommendation systems to make effective predictions without domain-specific data, ensuring they remain robust and relevant in diverse and evolving contexts. ZCDSR represents a more challenging setting compared to few-shot adaptation, as it requires the system to generalize effectively without any prior exposure to the target domain, fundamentally challenging the model's generalizability.

Existing works in zero-shot recommendation [5, 7, 12, 37] leverage metadata—such as text, images, and popularity—to transfer knowledge from source to unseen domains. The emergence of large language models (LLMs) has further advanced this direction by capturing rich item semantics and user intent through pretrained textual representations. Current LLM-based methods (LLM4Rec)



This work is licensed under a Creative Commons Attribution 4.0 International License. *RecSys '25, Prague, Czech Republic*
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1364-4/25/09
<https://doi.org/10.1145/3705328.3748077>

Table 1: Illustration of domain semantic bias in item descriptions from Industrial & Scientific (IS) vs. Video Games (VG).

Aspect	Industrial & Scientific (IS)	Video Games (VG)
Item Title	XXX Magnetics Magnet Fishing Kit	XXX Controller Faceplate
Example Description	"Includes powerful rare earth magnet with double braided polyester rope and carabiner. This strong magnet is ideal for ocean piers, lake docks, and bridges. Great for salvage and treasure hunting."	"Ultra fits for Xbox One X & One S controller; Completely fits flush on all sides; The shadow purple color looks great with a smooth grip, anti-slip, and sweat-free for long periods of gameplay."
Key Vocabulary	<i>magnet fishing, rare earth magnet, pull force, threadlocker, paracord</i>	<i>faceplate, side rails, anti-slip, soft touch, screw-driver</i>
Content Focus	Focused on durability, kit completeness, and outdoor usability, such as strength for fishing and treasure hunting.	Focused on controller modification, grip quality, and aesthetic customization for enhanced gaming experience.

can be categorized into: (1) *direct recommenders*[1, 16, 46], which generate predictions directly from textual inputs, and (2) *feature encoders*[19, 21, 29, 30], which generate semantic embeddings for downstream recommendation models.

However, when used as feature encoders, LLMs face a key challenge: **domain semantic bias**. This arises from mismatched vocabularies and content focus across domains, hindering the transferability of LLM-generated embeddings. For instance, as shown in Table 1, the Industrial & Scientific domain emphasizes technical attributes like *magnet strength* and *durability*, catering to *utility-focused users*, whereas Video Games focus on *visual design* and *grip comfort*, appealing to *gamers*. When embeddings trained in one domain are applied to another, the misaligned semantics can degrade recommendation performance. This highlights the need for methods that explicitly address cross-domain semantic misalignment to fully realize the zero-shot potential of LLMs.

This divergence in semantic focus highlights the domain-specific nature of item descriptions, which can significantly impact LLM-generated embeddings. When item embeddings trained on one domain are applied to another, the model may produce suboptimal recommendations due to the misalignment of feature representations. For instance, an LLM trained on the technical, performance-oriented descriptions of the Industrial & Scientific domain may fail to capture the subjective, experiential attributes emphasized in the Video Games domain. Such misalignment underscores the need for improved methods to bridge semantic gaps across domains, ensuring more robust and generalizable recommendation systems.

Our Insight: Domain-specific semantic bias exists in LLM-based zero-shot sequential recommenders. Therefore, it is crucial to strike a balance between generalization across domains and the preservation of domain-specific characteristics. Specifically, aligning item embeddings across domains while retaining the unique attributes of each domain is key to achieving effective cross-domain recommendation. Moreover, user behaviors tend to exhibit similarities across domains [28, 38], particularly in terms of the temporal or relational structures that govern item progression within a sequence, such as dynamic relationships and preference transitions. By capturing and leveraging these sequential patterns, we can enhance the recommendation process and make more accurate predictions, even in the absence of target-domain interaction data.

Present Work: We propose a novel model-agnostic LLM-based Recommendation Generalization framework for domain semantic

bias, **LLM-RecG**. LLM-RecG comprehensively addresses domain bias by capturing transferable sequential patterns while preserving domain-specific nuances, (e.g., distinct vocabularies, interaction behaviors, and content focuses), ensuring accurate recommendations in unseen target domains. Specifically, we introduce a novel training objective that balances *inter-domain compactness* and *intra-domain diversity* at the *item level*. *Inter-domain compactness* ensures that item embeddings are closely aligned across different domains, facilitating knowledge transfer and reducing domain-specific biases. In contrast, *intra-domain diversity* maintains the fine-grained distinctions among items within the same domain, ensuring the model does not overfit to dominant source-domain features." These two complementary objectives balance cross-domain alignment and domain-specific nuance, resulting in improved generalization and more accurate recommendations across domains. Furthermore, we transfer sequential patterns from the source domain to the target domain. Unlike item-level embedding generalization, sequential patterns capture how items are interacted with in specific sequences, providing critical context for user preferences. By clustering source user sequences into patterns, LLM-RecG learns to aggregate information from relevant sequential patterns through attention mechanisms during target domain inference. This enables dynamic adaptation of user embeddings without requiring target-domain interaction data, ensuring effective zero-shot recommendations. In summary, our main contributions are as follows:

Domain Semantic Bias-aware Framework: To the best of our knowledge, we are the **first** to identify and address the issue of domain semantic bias in LLM-based zero-shot cross-domain sequential recommendation (ZCDSR). Previous work typically focuses on leveraging domain-agnostic representations and overlooks the domain semantic bias, which significantly impact the transfer process. We, on the other hand, lay the groundwork for more effective knowledge transfer across domains by analyzing the sources of domain semantic bias and highlighting its effect on the performance of zero-shot recommendation tasks, which in turn leads to more accurate and reliable zero-shot predictions in unseen domains. Our results, both qualitative and quantitative, demonstrate that addressing this bias is crucial for ZCDSR and significantly enhances the model's generalizability.

Dual-level Generalization: We introduce a new generalization approach that enhances both *item-level* and *sequence-level* generalization. In contrast to prior methods that primarily focus on

static item embeddings or sequential patterns individually, these approaches often fail to holistically address both levels. Our approach combines item-level generalization via inter-domain compactness and intra-domain diversity with sequential pattern transfer across domains using attention mechanisms. This dual-level generalization results in more effective zero-shot recommendations, particularly in sparse or unseen domains, without relying on target-domain interaction data. Through extensive experiments, we show that LLM-RecG robust performance across domains, even with sparse and unseen target domains.

2 Problem Definition

In this section, we formally define the *zero-shot cross-domain sequential recommendation* (ZCDSR) task and introduce the notations used throughout this work.

Let $\mathcal{D}_s = \{\mathcal{V}_s, \mathcal{U}_s, \mathcal{X}_s\}$ represent the source domain, where $\mathcal{V}_s = \{v_1^s, v_2^s, \dots, v_{|\mathcal{V}_s|}^s\}$ denotes the set of items, and $\mathcal{U}_s = \{u_1^s, u_2^s, \dots, u_{|\mathcal{U}_s|}^s\}$ is the set of users in domain $\mathcal{D}_s$. Each item v_i^s is associated with metadata x_i^s , forming $\mathcal{X}_s$, the metadata set mapped one-to-one to the items. The goal of sequential recommendation in the source domain is to learn a scoring function that predicts the next item $v_{j,t}^s$ for a user u_j^s , given their interaction history $\mathcal{H}_j = \{v_{j,0}^s, v_{j,1}^s, \dots, v_{j,t-1}^s\}$. Formally, the scoring function is defined as: $\mathcal{F}(v_{j,t}^s | \mathcal{H}_j, \mathcal{X}_s)$, where $\mathcal{F}$ outputs a ranking score for candidate items based on the user's historical interactions and the item metadata.

Zero-shot Cross-Domain Sequential Recommendation:

Given a new target domain $\mathcal{D}_t = \{\mathcal{V}_t, \mathcal{U}_t, \mathcal{X}_t\}$, where $\mathcal{V}_t \cap \mathcal{V}_s = \emptyset$ and $\mathcal{U}_t \cap \mathcal{U}_s = \emptyset$, the objective is to produce a scoring function $\mathcal{F}'$ for $\mathcal{D}_t$ without training on $\mathcal{D}_t$ directly. This setting assumes no overlap in items or users between the source and target domains, making it a purely zero-shot transfer scenario.

We specifically focus on the zero-shot transfer setting rather than a few-shot setting, as it presents a more challenging yet crucial problem to study. Addressing this challenge is essential and fundamental for improving the scalability and adaptability of sequential recommendation systems across diverse domains.

3 Methodology

In this section, we present LLM-RecG, a model-agnostic generalization framework designed to address domain semantic bias in zero-shot cross-domain sequential recommendation (ZCDSR). Unlike few-shot adaptation, ZCDSR requires models to generalize to entirely unseen domains without any target-domain interactions. LLM-RecG tackles this challenge by enhancing the scalability and adaptability of sequential recommenders, enabling robust performance without further tuning.

At its core, LLM-RecG is built upon a semantic sequential framework (§ 3.1), which generalizes existing sequential recommenders to ZCDSR by incorporating an LLM-based semantic projection layer (§ 3.1.1). To further strengthen the generalization process, we propose a dual-level generalization strategy (§ 3.2 and § 3.3), which mitigates domain semantic discrepancies and improves the generalizability of the model in zero-shot scenarios.

3.1 Semantic Sequential Framework

The semantic sequential framework forms the backbone of LLM-RecG. It encodes item metadata into rich, high-dimensional embeddings and then mapping these embeddings into a space that captures sequential dependencies between items using any existing sequential recommenders, making it adaptable across domains. The use of an LLM-based semantic projection layer (§ 3.1.1) ensures that the model can process and integrate item-specific features in a way that captures the latent relationships between items in a sequence, allowing us to not only retain the domain semantic bias but also make it transferable to a target domain in a zero-shot setup.

3.1.1 LLM-based semantic projection layer. We leverage the power of LLMs to capture rich semantic information from item metadata. Specifically, we employ LLMs to extract semantic embeddings that encode contextual information from textual descriptions. For each item $v_i^s \in \mathcal{V}_s$ in the source domain, we aggregate its associated metadata (e.g., title, features, and description) into a unified textual description x_i^s , following a predefined template. We then adopt $\mathcal{E}$, a LLM-based semantic encoder to generate a high-dimensional semantic embedding $\mathbf{e}_i^{\text{sem}}$ of the item v_i^s :

$$\mathbf{e}_i^{\text{sem}} = \mathcal{E}(x_i^s), \quad (1)$$

where $\mathbf{e}_i^{\text{sem}} \in \mathbb{R}^{d_h}$ and d_h denotes the dimensionality of the semantic embedding space. Then, we apply a projection layer that maps the embeddings into a lower-dimensional latent space to reduce dimensionality and align the embeddings with user interaction patterns, enhancing the model's ability to capture sequential dependencies:

$$\mathbf{e}_i^s = \mathbf{W}_p \mathbf{e}_i^{\text{sem}} + \mathbf{b}_p, \quad (2)$$

where $\mathbf{e}_i^s \in \mathbb{R}^{d_l}$, $\mathbf{W}_p \in \mathbb{R}^{d_l \times d_h}$, and $\mathbf{b}_p \in \mathbb{R}^{d_l}$ are learnable parameters of the projection layer, with $d_l < d_h$.

3.1.2 Sequential Dependencies Modeling. Once the semantic embeddings for the items are obtained, we turn to sequential recommenders to capture the inherent sequential dependencies between items in a user's interaction history. We denote the sequence of low-dimensional embeddings corresponding to user u_j^s 's interaction history $\mathcal{H}_j^s = \{v_{j,0}^s, v_{j,1}^s, \dots, v_{j,|\mathcal{H}_j|}^s\}$ in the source domain as:

$$\mathbf{H}_j^s = \{\mathbf{e}_{j,0}^s, \mathbf{e}_{j,1}^s, \dots, \mathbf{e}_{j,|\mathcal{H}_j|}^s\}, \quad (3)$$

where $\mathbf{H}_j^s \in \mathbb{R}^{|\mathcal{H}_j^s| \times d_l}$ is the sequence of embeddings for user u_j^s in the source domain. This sequence is then passed to any sequential recommendation models *SeqRec*, forming the core of our model-agnostic approach. These models process the sequence and output a user representation:

$$\mathbf{y}_j^s = \text{SeqRec}(\mathbf{H}_j^s), \quad (4)$$

where $\mathbf{y}_j^s \in \mathbb{R}^{d_l}$ encodes the user's preferences based on their interaction history in the source domain.

To predict the relevance of candidate items, we compute a score function as follows:

$$\text{score}(h_j^s, v_i^s) = \langle \mathbf{y}_j^s, \mathbf{e}_i^s \rangle \quad (5)$$

where $\mathbf{e}_i^s \in \mathbb{R}^{d_l}$ is the embedding of the candidate item v_i^s in the source domain and $\langle \cdot \rangle$ denotes the dot product. The resulting score is then used to rank candidate items for recommendation.

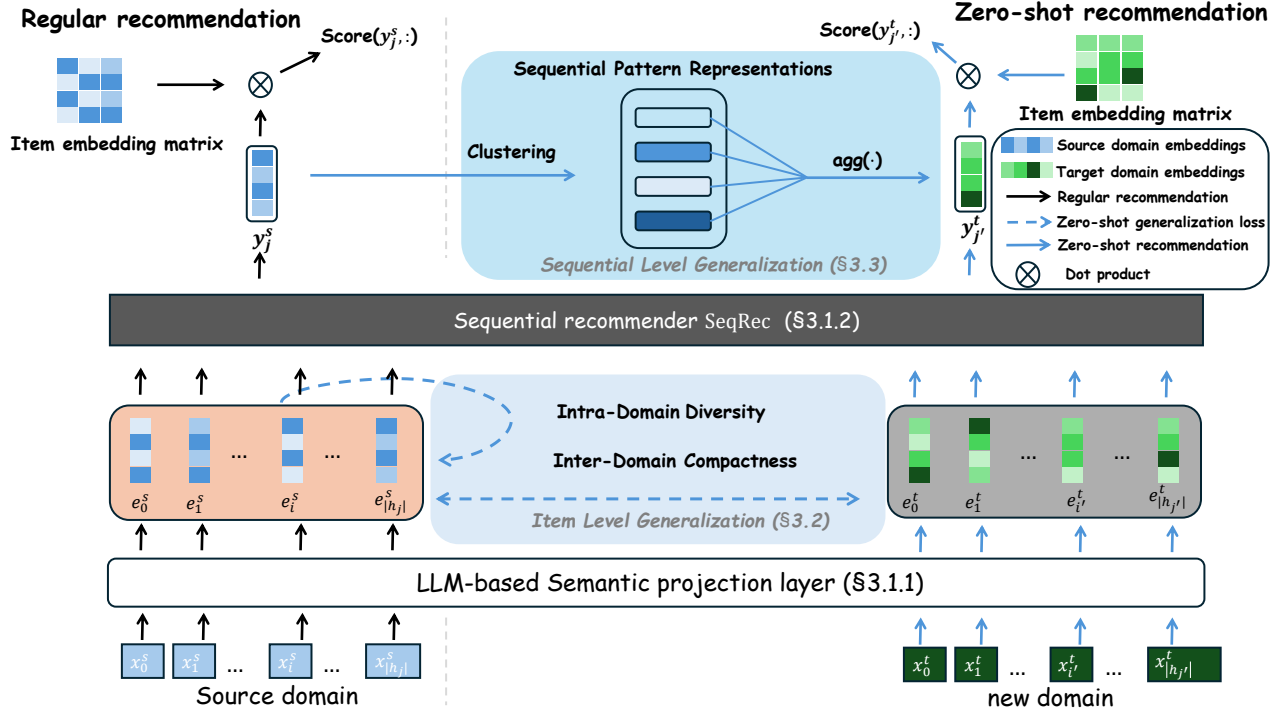


Figure 1: The model framework of LLM-RecG.

To optimize the model, we adopt the Bayesian Personalized Ranking (BPR) loss [31], which maximizes the pairwise ranking between positive and negative items. For a given user u_j^s in the source domain, the loss is defined as:

$$\mathcal{L}_{\text{rec}} = - \sum_j \sum_{(i^+, i^-)} \log \sigma \left(\text{score}(u_j^s, v_{i^+}^s) - \text{score}(u_j^s, v_{i^-}^s) \right), \quad (6)$$

where i^+ and i^- are positive and negative items, respectively, and $\sigma(\cdot)$ is the sigmoid function.

When performing zero-shot inference on a new domain, the same pretrained $\mathcal{E}$ and projection layers (i.e., $\mathbf{W}_p$ and $\mathbf{b}_p$) are used to process new items. For each item $v_{i'}^t \in \mathcal{V}^t$ in the target domain, the textual attributes $x_{i'}^t$ are transformed into semantic embeddings and subsequently projected into the low-dimensional space:

$$\mathbf{e}_{i'}^t = \mathbf{W}_p \mathbf{e}_{i'}^{\text{sem}} + \mathbf{b}_p. \quad (7)$$

This ensures that new item embeddings in the target domain are seamlessly integrated into the scoring framework without requiring model fine-tuning.

3.2 Item-Level Generalization

At the item level, we propose that semantic embeddings of items should satisfy the following two key properties:

Definition 3.1 (Inter-Domain Compactness). Inter-domain compactness ensures that item embeddings from different domains are closely aligned in the embedding space, reducing domain semantic biases and enabling effective knowledge transfer.

Definition 3.2 (Intra-Domain Diversity). Intra-domain diversity ensures that item embeddings within the same domain remain

distinct, capturing fine-grained variability and preventing representation collapse.

3.2.1 Inter-Domain Compactness. As defined, inter-domain compactness aligns item embeddings from different domains by minimizing the entropy of the similarity distribution between embeddings and the centers of other domains. Let $\{\mathbf{c}_d | d \in \{\mathcal{D}_s, \mathcal{D}_t\}\}$ represent the domain centers, and $\{\mathbf{e}_i^s, \mathbf{e}_{i'}^t | v_i^s \in \mathcal{V}_s, v_{i'}^t \in \mathcal{V}_t\}$ denote the item embeddings from the source and target domains.

The domain center $\mathbf{c}_d$ for a domain d ($d \in \{\mathcal{D}_s, \mathcal{D}_t\}$) is defined as the mean of all embeddings belonging to that domain:

$$\mathbf{c}_d = \frac{1}{|\mathcal{V}_d|} \sum_{v_{i'}^d \in \mathcal{V}_d} \mathbf{e}_{i'}^d, \quad (8)$$

where $\mathcal{V}_d$ represents the set of items in domain d , and $|\mathcal{V}_d|$ is the number of items in that domain.

To focus specifically on inter-domain alignment, the inter-domain compactness loss is formulated as:

$$\mathcal{L}_{\text{inter}} = \sum_{v_i \in \mathcal{V}} \sum_{\substack{d \in \{\mathcal{D}_s, \mathcal{D}_t\} \\ d \neq d_i}} Q_{id} \log Q_{id}, \quad (9)$$

where d_i indicates the domain of item v_i , and Q_{id} is the probability of embedding $\mathbf{e}_i^d$ being associated with domain d . This probability is computed as:

$$Q_{id} = \frac{\exp \left(\cos(\mathbf{e}_i^d, \mathbf{c}_d) / \tau \right)}{\sum_{\substack{d' \in \{\mathcal{D}_s, \mathcal{D}_t\} \\ d' \neq d_i}} \exp \left(\cos(\mathbf{e}_i^{d'}, \mathbf{c}_{d'}) / \tau \right)}, \quad (10)$$

where $\cos(\cdot)$ represents cosine similarity, and $\tau > 0$ is a temperature parameter that controls the sharpness of the similarity distribution.

Minimizing $\mathcal{L}_{\text{inter}}$ encourages embeddings to align with the centers of other domains, enhancing inter-domain compactness and reducing domain semantic biases. However, it alone is insufficient to ensure diversity within the same domain. To address this, we propose an additional objective for *intra-domain diversity*, which prevents embeddings from collapsing into overly similar representations and preserves fine-grained variability within each domain.

3.2.2 Intra-Domain Diversity. As defined, intra-domain diversity ensures distinctiveness among embeddings within the same domain, effectively capturing item-level variability and preventing representation collapse. For a domain $d \in \{\mathcal{D}_s, \mathcal{D}_t\}$ with item embeddings $\{\mathbf{e}_i^d | v_i^d \in \mathcal{V}_d\}$, the intra-domain diversity loss is formulated as:

$$\mathcal{L}_{\text{intra}} = - \sum_{d \in \{\mathcal{D}_s, \mathcal{D}_t\}} \frac{1}{|\mathcal{V}_d|} \sum_{v_i^d \in \mathcal{V}_d} \sum_{v_j^d \in \mathcal{V}_d} P_{ij} \log P_{ij}, \quad (11)$$

where P_{ij} denotes the similarity-based probability between items v_i^d and v_j^d within domain d . This probability is computed using cosine similarity with temperature scaling:

$$P_{ij} = \frac{\exp(\cos(\mathbf{e}_i^d, \mathbf{e}_j^d)/\tau)}{\sum_{v_k^d \in \mathcal{V}_d} \exp(\cos(\mathbf{e}_i^d, \mathbf{e}_k^d)/\tau)}, \quad (12)$$

where $\tau > 0$ regulates the sharpness of the similarity distribution. Lower τ sharpens the distribution, amplifying differences, while higher τ smooths it, promoting more uniform probabilities. Maximizing this entropy-based objective discourages embeddings within the same domain from collapsing into similar points, fostering richer intra-domain representation.

The generalization loss combines intra-domain diversity and inter-domain compactness:

$$\mathcal{L}_{\text{gen}} = -\alpha \mathcal{L}_{\text{intra}} + \beta \mathcal{L}_{\text{inter}}, \quad (13)$$

where $\alpha > 0$ and $\beta > 0$ control the balance between the two terms. This formulation promotes variability within domains while aligning embeddings across different domains, mitigating domain semantic biases. To balance the relative influence, β is scaled relative to α based on the number of domains $|\{\mathcal{D}_s, \mathcal{D}_t\}|$ and the total number of items $|N|$:

$$\beta = \alpha \frac{|N|}{|\{\mathcal{D}_s, \mathcal{D}_t\}|^3}. \quad (14)$$

This scaling ensures that inter-domain compactness contributes appropriately without overpowering intra-domain diversity.

To reduce computational complexity, we adopt a sampling strategy that approximates intra-domain and inter-domain components using batch data, significantly lowering overhead while enabling efficient integration with the recommendation loss.

The generalization loss $\mathcal{L}_{\text{gen}}$ complements the primary recommendation objective $\mathcal{L}_{\text{rec}}$ during training. The overall objective function is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{gen}}. \quad (15)$$

To preserve semantic information and enhance generalization, we introduce an additional projection layer. This layer maps semantic embeddings into another latent space, complementing the initial

projection. The final item embeddings are obtained by merging the outputs of both projection layers.

3.3 Sequence-level Generalization

Item-level generalization focuses on aligning static item embeddings across domains, but it overlooks the sequential dependencies in user behavior, which are crucial for understanding preferences and predicting future interactions. A sequential pattern abstracts the temporal or relational structure governing item progression within a sequence, capturing dynamic relationships and transitions in user preferences. These patterns highlight commonalities in user behavior across domains, making them particularly valuable for knowledge transfer. To bridge the gap left by item-level generalization, we propose leveraging these sequential patterns from the source domain to enhance user sequence representations during target domain inference.

3.3.1 Sequential Pattern Extraction from Source Domain. We extract *sequential patterns* that encapsulate common user behavioral trajectories by clustering the sequence embeddings of users from the source domain. Let $\mathbf{y}_j^s \in \mathbb{R}^{d_l}$ denote the sequence embedding of user u_j^s in the source domain, obtained by encoding the interaction history $h_j^s = \{v_1^s, v_2^s, \dots, v_{|h_j^s|}^s\}$.

The set of k sequential patterns $S = \{s_1, s_2, \dots, s_k\}$ is extracted by applying k -means clustering [27] over the sequence embeddings:

$$S = \text{k-means}(\{\mathbf{y}_j^s | u_j^s \in \mathcal{U}_s\}), \quad (16)$$

where $\mathcal{U}_s$ represents the set of users in the source domain. Each pattern $s_i \in \mathbb{R}^{d_l}$ corresponds to the centroid of a cluster, representing shared sequential behaviors observed in the source domain.

3.3.2 Soft Sequential Pattern Attention for Target Sequences. For zero-shot inference in the target domain, target user sequences are encoded to produce sequence embeddings $\mathbf{y}_j^t$ for user u_j^t , based on their interaction history $h_j^t = \{v_1^t, v_2^t, \dots, v_{|h_j^t|}^t\}$. Since target domain interaction data is unavailable during training, the model leverages source domain sequential patterns to guide the recommendation process. Rather than selecting the closest pattern directly, a soft attention mechanism is applied to aggregate information from multiple patterns.

The similarity between the target sequence embedding $\mathbf{y}_j^t$ and each sequential pattern s_i is computed using cosine similarity:

$$s_{j,i}^t = \cos(\mathbf{y}_j^t, s_i) = \frac{\mathbf{y}_j^t \cdot s_i}{\|\mathbf{y}_j^t\| \|s_i\|}. \quad (17)$$

These similarity scores are normalized using the softmax function to obtain attention weights over the sequential patterns:

$$\alpha_{j,i}^t = \frac{\exp(s_{j,i}^t)}{\sum_{l=1}^k \exp(s_{j,l}^t)}. \quad (18)$$

The attended sequential pattern representation $\tilde{s}_j^t$ for user u_j^t is computed as the weighted sum of the patterns:

$$\tilde{s}_j^t = \sum_{i=1}^k \alpha_{j,i}^t s_i. \quad (19)$$

Table 2: Processed Dataset Statistics.

Dataset	Items	Users	Avg. Seq. Len.
Industrial & Scientific (IS)	24,315	47,905	8.04
Video Games (VG)	24,030	89,746	8.60
Musical Instruments (MI)	23,618	55,787	8.88
Steam	8,156	26,541	19.68

3.3.3 Fusion of User Embedding and Pattern Representation. To integrate source domain patterns while preserving target-specific information, the target user embedding y_j^t is concatenated with the attended pattern representation $\tilde{s}_j^t$:

$$f_j^t = [y_j^t; \tilde{s}_j^t], \quad (20)$$

where $[\cdot; \cdot]$ denotes concatenation. The fused representation $f_j^t \in \mathbb{R}^{2d_l}$ is then projected back into the original embedding space:

$$g_j^t = W_f f_j^t, \quad (21)$$

where $W_f \in \mathbb{R}^{d_l \times 2d_l}$ is a learnable projection matrix. The resulting fused embedding g_j^t captures both the user’s target-specific preferences and the transferable sequential patterns from the source domain. It is subsequently used for next-item prediction.

4 Experiments

We conduct extensive experiments on three real-world datasets to evaluate the performance of LLM-RecG, following the settings outlined in § 3. To guide our analysis, we address the following research questions (RQs):

- **RQ1:** How well does LLM-RecG enhance baseline models in both in-domain and zero-shot scenarios?
- **RQ2:** What is the effectiveness of each component in LLM-RecG?
- **RQ3:** How does the key parameter impact performance?
- **RQ4:** How does LLM-RecG affect the alignment and uniformity of item embeddings across domains?

4.1 Experimental Setup

4.1.1 Datasets. We select three subsets i.e., *Video Games*, *Industrial & Scientific*, *Musical Instruments* from the Amazon Review dataset [14] with another cross-platform Steam Dataset [17], which contain user-item interactions and textual metadata for all items. Following the preprocessing outlined in previous works [11, 17, 20, 32], we filter out all users and items with fewer than ten interactions. Additionally, we exclude items that lack textual features, such as descriptions or attributes. For all datasets, the last item in each sequence is used for testing, while the penultimate item is used for validation. During evaluation, we adopt a negative sampling strategy, where the ground truth item is ranked against 100 randomly sampled negative items instead of the entire item set. The statistics of the processed datasets are summarized in Table 2.

4.1.2 Baseline & Evaluation Metrics. We present our method as a foundational framework and validate its effectiveness through comprehensive evaluations on three widely adopted recommendation models: **GRU4Rec** [13], **SASRec** [17], and **Bert4Rec** [34]. We denote variants enhanced with LLM-based semantic embeddings as -Sem and those further equipped with our proposed generalization loss as -RecG. Additionally, we also compared with two

state-of-the-art text-only cross-domain recommendation methods, **UniSRec** [15] and **RecFormer** [18]. For performance assessment, we adopt two widely used metrics: Recall **R@k** and Normalized Discounted Cumulative Gain (**N@k**).

4.1.3 Implementation Details. Our models are implemented using Python 3.9.20 and PyTorch 2.5.1¹. For optimization, we use the Adam optimizer across all models and adopt mini-batches with a batch size of 128. For LLM-based semantic encoder $\mathcal{E}$ mentioned in § 3.1, We adopt **LLM2Vec** [2] with Llama 3-8B [6]. To ensure fair comparisons, we perform a grid search to identify the optimal hyperparameters for each model. The search space includes embedding dimensionalities {64, 128, 256}, the number of layers {1, 2, 3}, dropout rates {0.2, 0.3, 0.5}, learning rates {0.001, 0.0005, 0.0001}, and, where applicable, the number of attention heads {1, 2, 4}, $\alpha \in \{0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001\}$. For evaluation, we use $k \in \{5, 10, 20\}$ for both R@k and N@k. Due to the page limitation, only $k = 10$ is reported. For all models, we repeat experiments 5 times with different random seeds and report the average performance.

4.2 Overall Comparison(RQ1)

4.2.1 ZCDJR performance. For the ZCDJR task, we follow the experimental settings outlined in Section 3.1. From Table 3, we observe the following key insights: a) Across all models and domain pairs, our method achieves substantial improvements over variants that directly use LLM-based semantic embeddings, with average gains exceeding 10% compared with best baselines even when transferring cross-platform between Amazon and Steam. This demonstrates the effectiveness of the proposed generalization framework in enhancing zero-shot recommendation performance while addressing domain semantic bias. b) Our method also improves robustness to domain shifts. For instance, GRU4Rec-Sem shows significant performance degradation depending on the relationship between source and target domains, with performance dropping by over 6% in R@10 in the VG domain when sourced from IS or MI. In contrast, our approach mitigates this variability, delivering more consistent and reliable results across diverse domain pairs. c) We find that models trained on the Steam dataset generally perform worst when transferred to other domains, likely due to semantic discrepancies in item descriptions. For example, BERT4Rec-Sem achieves only 9.97% at N@10 when transferring from Steam to IS—much lower than transfers from MI 17.06% or VG 17.67%. Despite this, our -RecG variants significantly improve performance under such domain shifts. In the same setting, BERT4Rec-RecG by relatively 28.9%, highlighting the effectiveness of our generalization loss in addressing semantic bias.

4.2.2 In-domain Sequential Recommendation Performance Comparison. In the in-domain scenario, we evaluate all baseline models, along with three base models each extended with two variants across four domains. The results in Table 4 yield several key observations: a) Variants augmented with LLM-based semantic information (-Sem) consistently outperform their corresponding base models, confirming the benefit of incorporating pretrained knowledge in domain-aligned tasks. For instance, BERT4Rec-Sem

¹Code and data available at: <https://github.com/yunzhe2/LLM-RecG>.

Table 3: Zero-shot performance across all source–target domain pairs. Throughout all result tables, we report Recall@10 (R@10) and NDCG@10 (N@10) in percentage (%), unless otherwise specified. Base models include two variants: **-Sem** (with LLM-based semantic embeddings) and **-RecG** (with the proposed generalization loss). Our method consistently outperforms all semantic-only variants across diverse domain pairs. It enhances robustness to domain shifts, reducing variability in performance across source–target combinations. Notably, even in challenging cross-platform transfers from Steam, **-RecG** delivers substantial gains (e.g., BERT4Rec-RecG improves N@10 by 28.9% over its **-Sem** counterpart when transferring to IS), effectively mitigating domain semantic bias.

Target	Source	GRU4Rec-Sem		SASRec-Sem		BERT4Rec-Sem		UniSRec		RecFormer		GRU4Rec-RecG		SASRec-RecG		BERT4Rec-RecG	
		R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10
IS	VG	20.02	9.69	30.27	16.51	27.22	14.57	27.29	14.62	<u>31.38</u>	<u>17.96</u>	27.94	15.47	36.94*	18.32*	35.23	17.67
	MI	21.78	10.48	29.88	16.68	29.94	16.31	29.67	16.12	<u>32.95</u>	<u>18.43</u>	28.97	16.39	38.59*	20.19*	37.21	20.10
	Steam	15.92	7.95	<u>23.22</u>	<u>12.94</u>	22.91	12.86	22.18	13.26	23.16	13.74	21.94	12.77	27.46	16.43	29.15*	16.82*
VG	IS	17.01	8.13	<u>32.97</u>	<u>18.68</u>	31.84	17.78	30.91	18.14	31.54	18.70	27.66	15.58	37.10*	22.59*	35.81	21.53
	MI	23.30	11.61	<u>34.73</u>	<u>19.46</u>	32.94	18.14	34.62	19.95	36.67	21.09	30.84	16.34	41.59*	23.67*	40.58	22.27
	Steam	16.19	7.74	<u>27.39</u>	<u>16.33</u>	26.17	15.52	26.49	14.85	27.54	16.74	23.73	12.17	29.46	17.49	29.26*	18.34*
MI	IS	16.35	7.81	26.20	<u>14.11</u>	24.82	13.28	25.16	13.85	27.69	17.29	23.89	12.54	32.16*	18.24*	30.10	17.06
	VG	20.08	9.75	<u>27.74</u>	<u>15.06</u>	25.13	13.32	26.91	15.62	28.44	17.67	26.83	14.18	34.42*	19.47*	32.51	18.81
	Steam	14.70	6.29	<u>20.46</u>	<u>11.83</u>	20.29	11.73	21.75	12.38	22.39	13.54	20.75	11.05	25.84	13.49	26.74*	13.92*
Steam	IS	16.24	7.62	<u>21.54</u>	<u>11.28</u>	19.75	9.97	20.36	11.36	21.18	11.83	21.44	12.37	23.59*	12.48*	22.49	12.85
	VG	18.52	7.80	<u>25.72</u>	<u>13.45</u>	20.53	11.62	21.52	11.74	23.06	12.47	22.61	12.59	29.26*	14.93*	25.11	13.79
	MI	17.73	7.75	<u>22.74</u>	<u>11.36</u>	19.13	9.41	21.86	12.01	23.29	12.31	22.87	12.14	26.13*	13.71*	22.72	12.97

Throughout all result tables, the best results are marked with * (t -test, $p \leq 0.05$) compared with the best baselines, unless otherwise specified.

Table 4: In-domain recommendation performance across four domains: IS, VG, MI, and Steam. Baseline models include GRU4Rec, SASRec, BERT4Rec, UniSRec, and RecFormer. Our **-Sem** variants consistently outperform base models, and **-RecG** achieves further significant gains by mitigating domain overfitting. Notably, **-RecG** outperforms all baselines across domains, demonstrating strong generalization and architecture-agnostic improvements.

Model	IS		VG		MI		Steam	
	R@10	N@10	R@10	N@10	R@10	N@10	R@10	N@10
GRU4Rec	36.05	22.02	54.32	34.52	40.75	25.23	54.61	32.70
SASRec	37.48	22.37	56.25	35.27	41.39	25.82	55.62	34.59
BERT4Rec	39.09	23.62	53.50	32.97	45.80	28.74	56.49	35.68
UniSRec	40.29	24.27	59.16	38.96	46.39	29.85	56.37	34.82
RecFormer	<u>42.36</u>	<u>25.16</u>	<u>61.49</u>	<u>40.28</u>	<u>51.24</u>	<u>31.65</u>	<u>58.17</u>	<u>36.92</u>
GRU4Rec-Sem	44.45	26.17	62.11	40.59	51.66	31.58	58.35	37.55
GRU4Rec-RecG	47.36	28.39	64.84	42.83	53.92	33.91*	59.62	38.14
SASRec-Sem	45.06	26.82	64.20	42.30	51.62	31.35	59.71	38.44*
SASRec-RecG	48.57*	29.11*	65.92*	43.74*	53.47	33.73	60.39	39.87
BERT4Rec-Sem	44.81	26.47	63.55	41.34	51.25	31.14	59.74	37.28
BERT4Rec-RecG	47.89	28.46	64.68	42.83	53.79*	32.96	61.63*	38.32

improves over BERT4Rec by up to 7.45% in R@10 (VG domain). b) Our proposed **-RecG** variants further outperform the **-Sem** models across all domains and models, with statistically significant gains. This highlights the effectiveness of our generalization loss in mitigating domain-specific overfitting due to the semantic gap between pretrained LLMs and the base recommendation models. c) The performance gains of **-RecG** over **-Sem** are especially notable for SASRec, a unidirectional architecture that benefits more from the added generalization signal. For example, SASRec-RecG improves over SASRec-Sem by 3.51% (R@10) and 2.91% (N@10) in the IS domain. d) Compared to state-of-the-art baselines (UniSRec and RecFormer), our **-RecG** variants achieve superior performance across all four domains. For instance, BERT4Rec-RecG achieves the highest R@10 in the MI and Steam domains with 53.79% and 61.63%,

respectively, outperforming RecFormer by 2.55% and 3.46%. Overall, these results demonstrate that our method not only improves in-domain effectiveness over traditional and text-only baselines but also provides a more general and robust framework applicable across different architectures.

Table 5: Ablation study on ZCDSR using BERT4Rec trained on IS and evaluated on VG and MI. Each component—item-level generalization (IG), intra-domain diversity (ID), inter-domain compactness (IC), and sequential-level generalization (SG)—contributes to performance. Removing ID causes the most significant degradation, highlighting its importance in preserving fine-grained domain distinctions. Overall, all modules collectively improve cross-domain generalization.

Models	VG		MI	
	R@10	N@10	R@10	N@10
BERT4Rec-RecG	35.81	21.53	30.10	17.06
w/o IG	32.49	18.36	26.49	14.83
w/o ID	28.26	16.33	20.93	12.17
w/o IC	32.74	19.79	27.35	15.77
w/o SG	33.17	19.22	28.63	16.52
BERT4Rec-Sem	31.84	17.78	24.82	13.28

4.3 Ablation Study(RQ2)

We conduct an ablation study to analyze the impact of each generalization module under the ZCDSR setting. Table 5 shows the performance of BERT4Rec on VG and MI with IS as the source domain. The results highlight the role of item-level and sequential-level generalization in improving ZCDSR performance. Below, we summarize the findings for each variant:

(a) *Remove Item level Generalization (IG).* When the item-level generalization is removed, the performance on both VG and MI drops noticeably (e.g., R@10 decreases from 35.81% to 32.49% on

VG and from 30.10% to 26.49% on MI). This demonstrates the importance of IG in aligning item embeddings across domains and improving transferability.

(b) *Remove Intra-domain Diversity (ID)*. Ablating intra-domain diversity (ID) causes the most significant performance degradation across all metrics, with R@10 dropping to 28.26% on VG and 20.93% on MI, even worse than -Sem variant. This emphasizes that maintaining fine-grained distinctions within each domain is crucial for preventing over-simplified representations, which can lead to poor generalization and reduced recommendation accuracy.

(c) *Remove Inter-domain Compactness (IC)*. Removing inter-domain compactness (IC) also results in a notable performance drop, particularly on VG (e.g., R@10 decreases from 35.81% to 32.74%). This indicates that aligning item embeddings across domains is essential for enabling effective knowledge transfer. However, the impact of removing IC is less severe than removing ID, suggesting that intra-domain diversity plays a more critical role in retaining domain-specific nuances.

(d) *Remove Sequential level Generalization (SG)*. Ablating sequential-level generalization (SG) reduces performance across both VG and MI, with R@10 dropping to 33.17% and 28.63%, respectively. This demonstrates that leveraging transferable sequential patterns is vital for capturing user behavior dynamics in the target domain. The smaller performance drop compared to IG and ID suggests that SG complements item-level generalization rather than acting as a standalone solution.

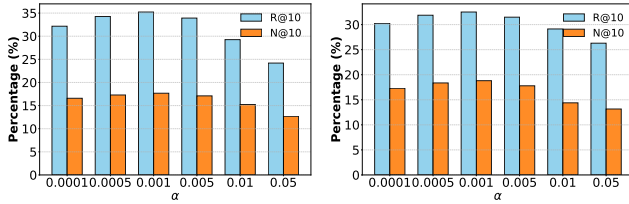


Figure 2: Impact of the generalization weight α on ZCDJR performance over IS (left) and MI (right) datasets. A moderate value of $\alpha = 0.001$ consistently yields the best performance, effectively balancing recommendation and generalization loss. Larger α values lead to performance degradation, highlighting the importance of tuning this parameter.

4.4 Sensitivity Analysis(RQ3)

To evaluate the impacts of key parameter α in ZCDJR tasks—we analyze its effect on the zero-shot performance of BERT4Rec-RecG across the IS and MI datasets. The hyperparameter α balances the recommendation loss and the generalization loss, making it a critical factor for optimizing ZCDJR performance. The results are visualized in Figure 2, and the following observations are made:

A moderate value of $\alpha = 0.001$ achieves the best performance across both IS and MI domains, with significant improvements in Recall@10 and NDCG@10. This suggests that $\alpha = 0.001$ effectively balances the recommendation loss, which ensures accurate predictions, and the generalization loss, which aligns item embeddings across domains. However, as α increases beyond 0.001,

performance declines consistently, indicating that overly large α values overemphasize the generalization loss, leading to suboptimal embeddings and reduced recommendation accuracy. The consistent trends across IS and MI further highlight α 's robustness and its importance as a key parameter for optimizing generalization in ZCDJR tasks.

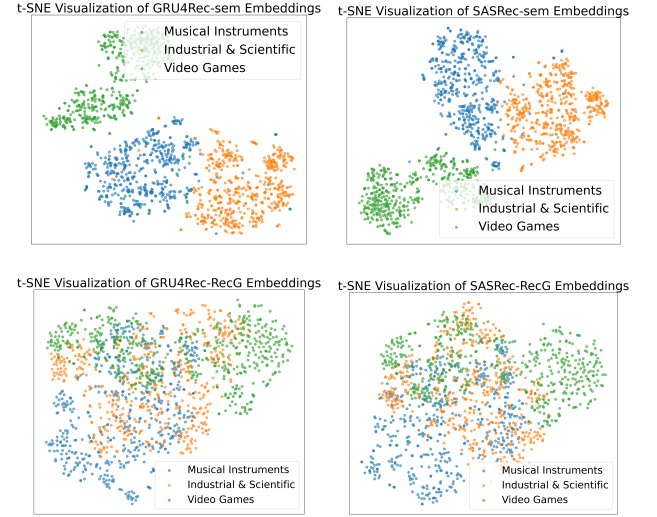


Figure 3: Visualization of item embeddings on the Video Games domain: (a) GRU4Rec-Sem, (b) SASRec-Sem, (c) GRU4Rec-RecG, and (d) SASRec-RecG. While -Sem variants produce semantically distinct clusters, -RecG yields more uniformly distributed embeddings, improving generalization and robustness across domains.

4.5 Visualization Analysis(RQ4)

To evaluate LLM-RecG affects the alignment and uniformity of item embeddings across domains—we analyze the item embeddings using t-SNE visualizations, as shown in Figure 3. These visualizations provide insights into the impact of our generalization framework on embedding alignment and generalization. The LLM-based semantic embeddings demonstrate a strong capability to distinguish items by leveraging prior knowledge. However, this strong semantic separation can hinder the generalization of the downstream recommender system. By incorporating the proposed generalization methods, the embeddings produced by the -RecG variants exhibit greater uniformity and are harder to distinguish across domains. This improved generalization enhances the model's robustness and adaptability.

5 Related Work

5.1 Cross-Domain Sequential recommendation

Unlike conventional sequential recommendation methods [17, 20, 34, 41] and cross-domain recommendation approaches [25, 33], cross-domain sequential recommendation (CDSR) requires models to leverage sequential dependencies to enhance recommendation performance across multiple domains. Early works, such as π -net [26] and other approaches [35, 39, 44], employ recurrent neural networks to capture sequential information among overlapping

users. More recently, C^2 DSR [3] introduces a contrastive Infomax loss with graph neural networks to capture both inter-sequence and intra-sequence item relationships. Unlike these methods, which assume full user overlap across domains, AMID [43] devises a multi-interest debiasing framework capable of handling both overlapping and non-overlapping users. In this paper, we address a more general challenge: how to effectively utilize user interactions and item metadata to improve recommendation performance in scenarios where there is no overlap between users or items across domains.

5.2 LLM-based Recommendation

LLMs' ability to generalize and understand text and long-term context has attracted interest in improving recommender systems. Research in this area is typically divided into generative and discriminative approaches based on LLMs' roles [42].

Generative recommender systems frame recommendation as natural language generation problems, enhancing interactivity and explainability. For instance, Rec Interpreter [45], ChatRec [9], and He et al. [12] use prompting strategies to convert user profiles or historical interactions into inputs for LLMs in conversational recommendations. ONCE [22] improves content-based recommendations by fine-tuning open-source LLMs and leveraging prompts with closed-source models. TALLRec [1] fine-tunes Alpaca [36] using self-instruct data to provide binary feedback ("yes" or "no"). GenRec [16] takes a different approach by directly generating the target item for recommendation. Additionally, CCF-LLM [24] encodes semantic and collaborative user-item interactions into hybrid prompts, fusing them via cross-modal attention for prediction.

Unlike generative systems, discriminative recommender systems do not require LLMs to respond during the inference stage, making them more practical and efficient. Early approaches, such as UniSRec [15], fine-tuned BERT [4] to associate item descriptions with transferable representations across recommendation scenarios. Harte et al. [10] improved sequential recommendation by replacing conventional model embeddings with LLM embeddings. LLM-ESR [23] further enhanced sequential recommendation by integrating LLM-generated semantic embeddings via retrieval-augmented self-distillation, avoiding additional inference costs. Additionally, [8, 40] leveraged LLMs to augment multi-domain recommendation models with scenario-specific knowledge, though their performance depends heavily on sample quality and domain expertise. In contrast, our method eliminates the reliance on domain knowledge, offering a more general and universal framework for cross-domain sequential recommendation.

6 Conclusion

This work addresses the challenge of domain semantic bias in LLM-based zero-shot cross-domain sequential recommendation (ZCDJR), a critical task for achieving accurate predictions in unseen domains. We propose LLM-RecG, a model-agnostic generalization framework that mitigates domain semantic bias by operating at both the item and sequential levels. At the item level, it balances inter-domain compactness with intra-domain diversity to align item embeddings while preserving domain-specific nuances. At the sequential level, it transfers user behavioral patterns through clustering and attention-based aggregation, enabling dynamic adaptation without requiring target-domain interaction data.

Comprehensive experiments show that LLM-RecG consistently improves ZCDJR performance across diverse domains compared to the state-of-the-art CDR baselines, validating its effectiveness and scalability. Future directions include integrating richer metadata and addressing cold-start users to further enhance personalized zero-shot recommendation.

References

- [1] Keqin Bao, Jizhi Zhang, Yang Zhang, Wenjie Wang, Fuli Feng, and Xiangnan He. 2023. Tallrec: An effective and efficient tuning framework to align large language model with recommendation. In *Proceedings of the 17th ACM Conference on Recommender Systems*. 1007–1014.
- [2] Parishad BehnamGhader, Vaibhav Adlakha, Marius Mosbach, Dzmitry Bahdanau, Nicolas Chapados, and Siva Reddy. 2024. LLM2Vec: Large Language Models Are Secretly Powerful Text Encoders. In *First Conference on Language Modeling*. <https://openreview.net/forum?id=IW1PR7vEBF>
- [3] Jiangxia Cao, Xin Cong, Jiawei Sheng, Tingwen Liu, and Bin Wang. 2022. Contrastive cross-domain sequential recommendation. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 138–147.
- [4] Jacob Devlin. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [5] Hao Ding, Yifei Ma, Anoop Deoras, Yuyang Wang, and Hao Wang. 2021. Zero-shot recommender systems. *arXiv preprint arXiv:2105.08318* (2021).
- [6] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783* (2024).
- [7] Philip J Feng, Pingjun Pan, Tingting Zhou, Hongxiang Chen, and Chuanjiang Luo. 2021. Zero shot on the cold-start problem: Model-agnostic interest learning for recommender systems. In *Proceedings of the 30th ACM international conference on information & knowledge management*. 474–483.
- [8] Zichuan Fu, Xiangyang Li, Chuhan Wu, Yichao Wang, Kuicai Dong, Xiangyu Zhao, Mengchen Zhao, Hui Feng Guo, and Ruiming Tang. 2023. A unified framework for multi-domain ctr prediction via large language models. *ACM Transactions on Information Systems* (2023).
- [9] Yunfan Gao, Tao Sheng, Youlin Xiang, Yun Xiong, Haofen Wang, and Jiawei Zhang. 2023. Chat-rec: Towards interactive and explainable llms-augmented recommender system. *arXiv preprint arXiv:2303.14524* (2023).
- [10] Jesse Harte, Wouter Zorgdrager, Panos Louridas, Asterios Katsifodimos, Dietmar Jannach, and Marios Fragkoulis. 2023. Leveraging large language models for sequential recommendation. In *Proceedings of the 17th ACM Conference on Recommender Systems*. 1096–1102.
- [11] Ruining He and Julian McAuley. 2016. Fusing similarity models with markov chains for sparse sequential recommendation. In *2016 IEEE 16th international conference on data mining (ICDM)*. IEEE, 191–200.
- [12] Zhankui He, Zhouhang Xie, Rahul Jha, Harald Steck, Dawen Liang, Yesu Feng, Bodhisattwa Prasad Majumder, Nathan Kallus, and Julian McAuley. 2023. Large language models as zero-shot conversational recommenders. In *Proceedings of the 32nd ACM international conference on information and knowledge management*. 720–730.
- [13] B Hidas. 2015. Session-based Recommendations with Recurrent Neural Networks. *arXiv preprint arXiv:1511.06939* (2015).
- [14] Yupeng Hou, Jiacheng Li, Zhankui He, An Yan, Xiusi Chen, and Julian McAuley. 2024. Bridging Language and Items for Retrieval and Recommendation. *arXiv preprint arXiv:2403.03952* (2024).
- [15] Yupeng Hou, Shanlei Mu, Wayne Xin Zhao, Yaliang Li, Bolin Ding, and Ji-Rong Wen. 2022. Towards universal sequence representation learning for recommender systems. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 585–593.
- [16] Jianchao Ji, Zelong Li, Shuyuan Xu, Wenyue Hua, Yingqiang Ge, Juntao Tan, and Yongfeng Zhang. 2024. Genrec: Large language model for generative recommendation. In *European Conference on Information Retrieval*. Springer, 494–502.
- [17] Wang-Cheng Kang and Julian McAuley. 2018. Self-attentive sequential recommendation. In *2018 IEEE international conference on data mining (ICDM)*. IEEE, 197–206.
- [18] Jiacheng Li, Ming Wang, Jin Li, Jinmiao Fu, Xin Shen, Jingbo Shang, and Julian McAuley. 2023. Text is all you need: Learning language representations for sequential recommendation. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 1258–1267.
- [19] Ruyi Li, Wenhao Deng, Yu Cheng, Zheng Yuan, Jiaqi Zhang, and Fajie Yuan. 2023. Exploring the upper limits of text-based collaborative filtering using large language models: Discoveries and insights. *arXiv preprint arXiv:2305.11700* (2023).
- [20] Yunzhe Li, Yue Ding, Bo Chen, Xin Xin, Yule Wang, Yuxiang Shi, Ruiming Tang, and Dong Wang. 2021. Extracting Attentive Social Temporal Excitation for Sequential Recommendation. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*. 998–1007.

- [21] Youhua Li, Hanwen Du, Yongxin Ni, Pengpeng Zhao, Qi Guo, Fajie Yuan, and Xiaofang Zhou. 2024. Multi-modality is all you need for transferable recommender systems. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)*. IEEE, 5008–5021.
- [22] Qijiong Liu, Nuo Chen, Tetsuya Sakai, and Xiao-Ming Wu. 2024. Once: Boosting content-based recommendation with both open-and closed-source large language models. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*. 452–461.
- [23] Qidong Liu, Xian Wu, Yejing Wang, Zijian Zhang, Feng Tian, Yefeng Zheng, and Xiangyu Zhao. [n. d.]. LLM-ESR: Large Language Models Enhancement for Long-tailed Sequential Recommendation. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.
- [24] Zhongzhou Liu, Hao Zhang, Kuicai Dong, and Yuan Fang. 2024. Collaborative Cross-modal Fusion with Large Language Model for Recommendation. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 1565–1574.
- [25] Linhao Luo, Yumeng Li, Buyu Gao, Shuai Tang, Sinan Wang, Jiancheng Li, Tan-chao Zhu, Jiancai Liu, Zhao Li, and Shirui Pan. 2023. MAMDR: A model agnostic learning framework for multi-domain recommendation. In *2023 IEEE 39th International Conference on Data Engineering (ICDE)*. IEEE, 3079–3092.
- [26] Muyang Ma, Pengjie Ren, Yujie Lin, Zhumin Chen, Jun Ma, and Maarten de Rijke. 2019. π -net: A parallel information-sharing network for shared-account cross-domain sequential recommendations. In *Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval*. 685–694.
- [27] James MacQueen et al. 1967. Some methods for classification and analysis of multivariate observations. In *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability*, Vol. 1. Oakland, CA, USA, 281–297.
- [28] Vilfredo Pareto. 1897. *Cursus de l'économie politique*. F. Rouge, Lausanne.
- [29] Zhaopeng Qiu, Xian Wu, Jingyue Gao, and Wei Fan. 2021. U-BERT: Pre-training user representations for improved recommendation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 4320–4327.
- [30] Shashank Rajput, Nikhil Mehta, Anima Singh, Raghunandan Hulikal Keshavan, Trung Vu, Lukasz Heldt, Lichan Hong, Yi Tay, Vinh Tran, Jonah Samost, et al. 2023. Recommender systems with generative retrieval. *Advances in Neural Information Processing Systems* 36 (2023), 10299–10315.
- [31] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2012. BPR: Bayesian personalized ranking from implicit feedback. *arXiv preprint arXiv:1205.2618* (2012).
- [32] Steffen Rendle, Christoph Freudenthaler, and Lars Schmidt-Thieme. 2010. Factorizing personalized markov chains for next-basket recommendation. In *Proceedings of the 19th international conference on World wide web*. 811–820.
- [33] Xiang-Rong Sheng, Liqin Zhao, Guorui Zhou, Xinyao Ding, Binding Dai, Qiang Luo, Siran Yang, Jingshan Lv, Chi Zhang, Hongbo Deng, et al. 2021. One model to serve all: Star topology adaptive recommender for multi-domain ctr prediction. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*. 4104–4113.
- [34] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. 2019. BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In *Proceedings of the 28th ACM international conference on information and knowledge management*. 1441–1450.
- [35] Wenchao Sun, Muyang Ma, Pengjie Ren, Yujie Lin, Zhumin Chen, Zhaochun Ren, Jun Ma, and Maarten De Rijke. 2021. Parallel split-join networks for shared account cross-domain sequential recommendations. *IEEE Transactions on Knowledge and Data Engineering* 35, 4 (2021), 4106–4123.
- [36] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. 2023. Stanford alpaca: An instruction-following llama model.
- [37] Juntao Wang, Adit Krishnan, Hari Sundaram, and Yunzhe Li. 2023. Pre-trained Neural Recommenders: A Transferable Zero-Shot Framework for Recommendation Systems. *arXiv preprint arXiv:2309.01188* (2023).
- [38] Juntao Wang, Praneet Rathi, and Hari Sundaram. 2024. A Pre-trained Zero-shot Sequential Recommendation Framework via Popularity Dynamics. In *18th ACM Conference on Recommender Systems (RecSys '24)*. ACM, 433–443. doi:10.1145/3640457.3688145
- [39] Yaqing Wang, Caili Guo, Yunfei Chu, Jenq-Neng Hwang, and Chunyan Feng. 2020. A cross-domain hierarchical recurrent model for personalized session-based recommendations. *Neurocomputing* 380 (2020), 271–284. <https://api.semanticscholar.org/CorpusID:209903109>
- [40] Yuhao Wang, Yichao Wang, Zichuan Fu, Xiangyang Li, Wanyu Wang, Yuyang Ye, Xiangyu Zhao, Huifeng Guo, and Ruiming Tang. 2024. Llm4msr: An llm-enhanced paradigm for multi-scenario recommendation. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 2472–2481.
- [41] Yule Wang, Xin Xin, Yue Ding, Yunzhe Li, and Dong Wang. 2021. ICPE: An item cluster-wise pareto-efficient framework for recommendation debiasing. *arXiv preprint arXiv:2109.12887* (2021).
- [42] Likang Wu, Zhi Zheng, Zhaopeng Qiu, Hao Wang, Hongchao Gu, Tingjia Shen, Chuan Qin, Chen Zhu, Hengshu Zhu, Qi Liu, et al. 2024. A survey on large language models for recommendation. *World Wide Web* 27, 5 (2024), 60.
- [43] Wujiang Xu, Qitian Wu, Runzhong Wang, Mingming Ha, Qiongxi Ma, Linxun Chen, Bing Han, and Junchi Yan. 2024. Rethinking cross-domain sequential recommendation under open-world assumptions. In *Proceedings of the ACM on Web Conference 2024*. 3173–3184.
- [44] Guang Yang, Xiaoguang Hong, Zhaohui Peng, and Yang Xu. 2020. Long Short-Term Memory with Sequence Completion for Cross-Domain Sequential Recommendation. In *APWeb/WAIM*. <https://api.semanticscholar.org/CorpusID:224770681>
- [45] Zhengyi Yang, Jiancan Wu, Yanchen Luo, Jizhi Zhang, Yancheng Yuan, An Zhang, Xiang Wang, and Xiangnan He. 2023. Large language model can interpret latent space of sequential recommender. *arXiv preprint arXiv:2310.20487* (2023).
- [46] Zhenrui Yue, Sara Rabhi, Gabriel de Souza Pereira Moreira, Dong Wang, and Even Oldridge. 2023. LlamaRec: Two-stage recommendation using large language models for ranking. *arXiv preprint arXiv:2311.02089* (2023).